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Ultrasound Methods and Systems for Measuring Physiological Properties

Abstract

Ultrasound methods and systems for measuring physiological properties are disclosed. The ultrasound methods and systems measure one or more characteristics of a vessel, such as vessel-wall displacement over time or blood-flow velocity over time, based on a pulse wave propagating through the vessel. In aspects, the characteristics are measured at two locations of the same vessel with a known distance between the two locations. A time shift between the measured characteristics at the two locations is calculated and used, along with the known distance, to determine one or more physiological properties, such as pulse-wave velocity or blood pressure. These physiological properties can be measured without the assistance of ECG data.

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Background/Summary

BACKGROUND

[0001] Ultrasound is widely used to image human cardiac anatomy and detect cardiac functions, such as ejection fraction and left ventricle outflow traction (LVOT). Conventional ultrasound lacks the capability, however, to accurately measure other physiological properties, such as pulse-wave velocity (PWV) and blood pressure (BP). Some of the challenges of measuring pulse-wave velocity include, for example, the fact that pulse-wave velocity (e.g., the speed of a pulse wave propagation through the blood stream) is extremely fast, which causes conventional systems to be unable to directly measure the pulse wave. Blood pressure is also challenging to measure with ultrasound because blood pressure depends on several physiological parameters including, for example, vessel diameter, vessel stiffness, cardiac output pressure, and distance between the heart and a measurement location. Even with the assistance of an electrocardiogram (ECG) to monitor a heart cycle, measuring PWV is difficult due to the long distance between the heart and the measurement location.

SUMMARY

[0002] Ultrasound methods and systems for measuring physiological properties are disclosed. The ultrasound methods and systems can measure one or more characteristics of a vessel, such as vessel-wall displacement over time or blood-flow velocity over time, based on a pulse wave propagating through the vessel. In aspects, the characteristics are measured at two locations of the same vessel with a known distance between the two locations. A time shift between the measured characteristics at the two locations is calculated and used, along with the known distance, to determine one or more physiological properties, such as pulse-wave velocity and/or blood pressure. These physiological properties can be measured without the assistance of ECG data.

[0003] In some aspects, an ultrasound system is disclosed. The ultrasound system can include an ultrasound scanner to generate ultrasound data based on reflections of ultrasound signals transmitted by the ultrasound scanner at an anatomy. The ultrasound scanner is also configured to acquire the ultrasound data at two locations of a same vessel that are separated by a distance, and the ultrasound data is acquired at the two locations within a time interval. The ultrasound system also includes one or more processors and one or more computer-readable storage media having instructions stored thereon that, responsive to execution by the one or more processors, cause the one or more processors to: determine one or more characteristics of the vessel at each of the two locations using the ultrasound data; determine, based on a correlation between the one or more characteristics of the vessel at the two locations, a phase delay of a pulse wave propagating through the vessel between the two locations; and determine a physiological property associated with the vessel based on the one or more characteristics and the known distance.

[0004] In aspects, a method is disclosed. The method includes acquiring ultrasound data at two locations of a same vessel that are separated by a known distance, the ultrasound data acquired at the two locations within a time interval. The method also includes determining one or more characteristics of the vessel at each of the two locations using the ultrasound data. In addition, the method includes determining, based on a correlation between the one or more characteristics of the vessel at the two locations, a phase delay of a pulse wave propagating through the vessel between the two locations. The method further includes determining a physiological property associated with the vessel based on the one or more characteristics and the known distance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The appended drawings illustrate examples and are, therefore, exemplary embodiments and not considered to be limiting in scope. Throughout the drawings, the same numbers are used to

reference like features and components.
[0006] FIG. **1** illustrates an example environment for an ultrasound system having an ultrasound scanner, in accordance with one or more implementations.
[0007] FIG. **2** illustrates an example implementation of the ultrasound system from FIG. **1**.
[0008] FIG. **3** illustrates an example implementation of a multi-mode ultrasound in accordance with one or more implementations.
[0009] FIG. **4** illustrates an example implementation of measuring PWV using dual M-mode ultrasound.
[0010] FIG. **5** illustrates an example implementation of measuring PWV using dual-pulsed-wave (PW) Doppler.
[0011] FIG. **6** illustrates an example implementation of using dual-PW Doppler to measure both PWV and vessel-wall movement.
[0012] FIG. **7** illustrates an example implementation of using a combination mode to measure PWV.
[0013] FIG. **8** depicts a method for using ultrasound to measure physiological properties.
[0014] FIG. **9** depicts a method for measuring physiological properties using ultrasound.
[0015] FIG. **10** represents an example machine-learning (ML) model for processing an input from an ultrasound device.
[0016] FIG. **11** represents an example machine-learning architecture used to train an ML model.
[0017] FIG. **12** represents an example model using a convolutional neural network (CNN) to process an input image, which can include representations of objects that can be identified via object recognition.
[0018] FIG. **13** illustrates an example user interface displayed via a display device coupled to the ultrasound system from FIG. **1**, in accordance with one or more implementations.

DETAILED DESCRIPTION

[0019] Disclosed herein are ultrasound methods and systems for measuring physiological properties. These techniques and systems provide increased accuracy compared to conventional techniques used to measure physiological properties. The techniques and systems disclosed herein use ultrasound, without the assistance of an electrocardiogram (ECG), to measure such physiological properties of a subject, which reduces errors in the calculations that are typically due to estimations of the distance between the heart and the actual location being measured. Example physiological properties being measured include pulse-wave velocity and blood pressure.
[0020] Pulse-wave velocity (PWV) is the speed at which pressure waves move through the circulatory system, such as an artery or a combined length of arteries or blood vessels. PWV is an independent predictor of cardiovascular risk. Assessment of PWV can be performed noninvasively by measuring the carotid and femoral pulse pressures and the time delay between the two. PWV increases with increased arterial stiffness. In addition, PWV is a reliable prognostic marker for cardiovascular morbidity and mortality. Research shows that hypertension contributes to an increase in age-related arterial stiffening. While blood pressure (BP) is a valuable first-level indicator of hypertension, PWV provides further detail. The mean values of pulse-wave velocity for 20-29 year olds is 5.8±0.7 meters per second (m/s) and for 40-49 years is 7.1±0.9 m/s. Because BP is one of the most important contributing factors to PWV, measuring the pulse-wave velocity (PWV) is generally considered to be a promising technique for continuous noninvasive measurements of BP. However, conventional methods to measure PWV are not very accurate or convenient. For example, to measure PWV conventionally, an ECG-aided ultrasound method was used. However, in such a method, the distance between the heart and the measurement location is estimated and can lead to a huge error.
[0021] The techniques and systems disclosed herein use only an ultrasound system (without using ECG) to measure PWV and BP. Using these techniques, distance estimation errors are reduced or eliminated, resulting in higher accuracy of the measurement compared to conventional techniques.

The pulse wave is a pressure wave initialized by the heart and propagates along the vessels, which leads to two consequences: first, the vessel wall is compressed and the compression ratio depends on the vessel wall stiffness; second, the blood flow changes due to the pressure wave. Accordingly, different methods can be used to measure PWV. One method disclosed herein measures the rate of change of a vessel diameter (vessel-diameter-change speed), which is the basis for a dual M-mode method. Another method disclosed herein measures changes in blood-flow velocity (blood-flow-change speed), which is the basis for a dual-PW method. Yet another method disclosed herein is a combination of measuring the vessel-diameter-change speed and the blood-flow-change speed, which is the basis for a combination M-mode and PW method. Further details of these methods and other features are described below.

Example Ultrasound System

[0022] FIG. **1** illustrates an example environment for an ultrasound system **100** having an ultrasound scanner, in accordance with one or more implementations. Generally, the ultrasound system **100** includes an ultrasound machine **102**, which generates data based on high-frequency sound waves reflecting off body structures. The ultrasound machine **102** includes various components, some of which include a scanner **104**, one or more processors **106**, a display device **108**, and a memory **110**.

[0023] A user **112** (e.g., nurse, ultrasound technician, clinician, operator, sonographer) directs the scanner **104** toward a patient **114** (e.g., a human or animal) to non-invasively scan internal bodily structures (anatomies, organs, tissues, etc.) of the patient **114** for testing, diagnostic, or therapeutic reasons. In some implementations, the scanner **104** includes an ultrasound transducer array and electronics communicatively coupled to the ultrasound transducer array to transmit ultrasound signals to the patient's anatomy and receive ultrasound signals reflected from the patient's anatomy (e.g., echoes) as part of an ultrasound examination. In some implementations, the scanner **104** is an ultrasound scanner, which can also be referred to as an ultrasound probe or transducer.

[0024] The display device **108** is coupled to the processor **106**, which processes the reflected ultrasound signals to generate ultrasound data. The display device **108** is configured to generate and display an ultrasound image (e.g., ultrasound image **116**) of the anatomy based on the ultrasound data generated by the processor **106** from the reflected ultrasound signals detected by the scanner **104**. In some aspects, the ultrasound data includes the ultrasound image **116** or data representing the ultrasound image **116**.

[0025] FIG. **2** illustrates an example implementation **200** of the ultrasound system **100** from FIG. **1**. The scanner **104** (e.g., ultrasound scanner) includes an enclosure **202** extending between a distal end portion **204** and a proximal end portion **206**. The enclosure **202** includes a central axis **208** (e.g., longitudinal axis) that intersects the distal end portion **204** and the proximal end portion **206**. The central axis **208** corresponds to an axial direction of the scanner **104**. In an example, the scanner **104** is electrically coupled to an ultrasound imaging system (e.g., the ultrasound machine **102**) via a cable **210** that is attached to the proximal end portion **206** of the scanner **104** by a strain-relief element **212**. In some implementations, the scanner **104** is wirelessly coupled to the ultrasound imaging system and communicates with the ultrasound imaging system via one or more wireless transmitters, receivers, or transceivers over a wireless connection or network (Bluetooth™, Wi-Fi™, etc.).

[0026] A transducer assembly **214** having one or more transducer elements is electrically coupled to system electronics **216** in the ultrasound machine **102**. In operation, the transducer assembly **214** transmits ultrasound energy from the one or more transducer elements toward a subject and receives ultrasound echoes from the subject. The ultrasound echoes are converted into electrical signals by the transducer element(s) and electrically transmitted to the system electronics **216** in the ultrasound machine **102** for processing and generation of one or more ultrasound images.

[0027] Capturing ultrasound data from a subject using a transducer assembly (e.g., the transducer assembly **214**) generally includes generating ultrasound signals, transmitting ultrasound signals into the subject, and receiving ultrasound signals reflected by the subject. A wide range of frequencies of ultrasound can be used to capture ultrasound data, such as, for example, low-frequency ultrasound (e.g., less than 15 megahertz (MHz)) and/or high-frequency ultrasound (e.g., greater than or equal to 15 MHz). A particular frequency range to use can readily be determined based on various factors, including, for example, depth of imaging, desired resolution, and so forth.

[0028] In some implementations, the system electronics **216** include one or more processors (e.g., the processor(s) **106** from FIG. **1**), integrated circuits, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), Graphics Processing Units (GPUs) and power sources to support functioning of the ultrasound machine **102**. In some implementations, the ultrasound machine **102** also includes an ultrasound control subsystem **218** having one or more processors. At least one processor, FPGA, ASIC, or GPU causes electrical signals to be transmitted to the transducer(s) of the scanner **104** to emit sound waves and also receives electrical pulses from the scanner **104** that were created from the returning echoes. One or more processors, FPGAs, ASICs, or GPUs process the raw data associated with the received electrical pulses and form an image that is sent to an ultrasound imaging subsystem **220**, which causes the image (e.g., the image **116** in FIG. **1**) to be displayed via the display device **108**. Thus, the display device **108** displays ultrasound images from the ultrasound data processed by the processor(s) of the ultrasound control subsystem **218**.

[0029] In some implementations, the ultrasound machine **102** also includes one or more user input devices (a keyboard, a cursor control device, a microphone, a camera, etc.) that input data and enable taking measurements from the display device **108** of the ultrasound machine **102**. The ultrasound machine **102** can also include a disk storage device (e.g., computer-readable storage media such as read-only memory (ROM), a Flash memory, a dynamic random-access memory (DRAM), a NOR memory, a static random-access memory (SRAM), a NAND memory, and so on) for storing the acquired ultrasound data. In aspects, the disk storage device includes the memory **110**, which is local to the ultrasound machine **102**. Alternatively, the memory **110** used for storing the acquisition data can be remote, such as on a remote server communicatively connected to the ultrasound machine **102**. In addition, the ultrasound machine **102** can include a printer that prints the image from the displayed data. To avoid obscuring the techniques described herein, such user input devices, disk storage device, and printer are not shown in FIG. **2**.

[0030] FIG. **3** illustrates an example implementation **300** of a multi-mode ultrasound **302** in accordance with one or more implementations. The ultrasound modes illustrated and described with respect to FIG. **3** are described as examples and the implementations disclosed herein are not intended to be limited to these examples. The techniques disclosed herein can be implemented with other ultrasound modes that are not described herein. Accordingly, the list of example ultrasound modes is not an exhaustive list.

[0031] The multi-mode ultrasound **302** includes various combinations of modes utilized by the ultrasound control subsystem **218** to process raw data associated with received signals from the transducer(s). In implementations, the ultrasound control subsystem **218** can use multiple ultrasound modes to process the raw ultrasound data to measure physiological properties. Such a multi-mode operation can include a dual implementation of a particular mode, including M-mode, pulsed wave (PW) Doppler (also referred to herein as "PW"), etc. For example, the multi-mode ultrasound **302** can include a dual M-mode **304**, a dual PW **306**, or a combination mode **308**. The combination mode is a combination of two or more different modes, including M-mode+PW, a combination of B-mode+M-mode+PW Doppler, a combination of B-mode+M-mode+color Doppler+PW Doppler, etc. In implementations, the user can selectively choose which combination of modes the ultrasound system uses. If, for example, a high frame rate is desired, then the user may select the dual M-mode **304**, the dual PW **306**, or a combination mode including M-mode and PW Doppler used simultaneously.

[0032] In some aspects, the ultrasound machine **102** uses a first mode (e.g., current mode **310**), which can be selected by the user or initialized by a setting. Using the current setting **310**, the ultrasound machine **102** generates the ultrasound image **126**. The ultrasound image **126** is sent to the processor(s) **106**, which uses the ultrasound image **126** to determine one or more factors, including image quality, protocol step, neural-network inference, etc. In one example, the processor(s) **106** can receive an inference from a neural network using the ultrasound image **126** as an input. In addition, the processor(s) **106** can determine a confidence factor associated with the one or more factors, such as a weight assigned to each factor. The processor(s) **106** can use, as input for determining the one or more factors and the confidence factor, various data from the database **130**, including protocol data, user data, patient history, previous measurements of BP and/or PWV, etc.

[0033] The processor(s) **106** can use the one or more factors and the confidence factor to determine a score **312** for the ultrasound image **126**. The score **312** is provided to a controller **314**. The controller **314** can use the score **312** to select a mode of the multi-mode ultrasound **302** for use in generating subsequent ultrasound images. The controller **314** can generate a mode selection **316** for the multi-mode ultrasound **302**. The mode selection **316** can select the same mode as the current mode **310**. In aspects, the mode selection **316** can select a mode that is different from the current mode **310** to enable the ultrasound machine **102** to generate an improved ultrasound image having, for example, a higher image quality for obtaining more-accurate measurements.

Dual M-mode

[0034] FIG. **4** illustrates an example implementation of measuring PWV using a dual M-mode ultrasound (e.g., dual M-mode **304** in FIG. **3**). In this example, to measure PWV, an ultrasound scanner applies two M-lines on the same blood vessel (e.g., an artery) with a known distance between the two M-lines. For example, setup **400** shows one or more ultrasound scanner(s) **402** (e.g., scanner **104**) applying two M-mode lines (e.g., M-line-1 **404** and M-line-2 **406**) at two locations on a vessel **408** that are separated by a distance d.sub.1 **410**. The distance d.sub.1 **410** is known (e.g., predefined), such as a defined distance between the two M-lines. The two M-lines can be applied by the same scanner or two separate scanners. When applied by a single scanner, the distance d.sub.1 **410** is small (5 mm, 2 mm, between 1 and 0.001 mm, etc.), such that the two M-lines are applied without physically moving the scanner. The distance d.sub.1 **410** can be determined based on the transducer elements used to generate the two M-lines. For example, the transducer elements used to generate the M-lines are separated by a fixed distance, which is equal to the distance d.sub.1 **410**. The distance d.sub.1 **410** can be greater when applied by separate scanners; however, a greater distance d.sub.1 **410** may be less accurately defined and can therefore reduce the accuracy of the PWV calculations. When applied by separate scanners, each scanner can include a position-measuring sensor, such as an internal measurement unit (IMU) that measures positional data of the scanner. The positional data can include an acceleration, an angular rate, and/or an orientation of the scanner. The distance d.sub.1 **410** can be estimated based on the positional data generated by the IMU of each scanner and the firing of the ultrasound transducer array of each scanner.

[0035] Due to the pulse wave passing through the vessel **408** and based on a stiffness (or flexibility) of the vessel, the vessel wall moves, resulting in changes to the diameter of the vessel. The rate of movement of the vessel wall, or the rate of change of the vessel diameter, is related to the PWV. Using the dual M-mode **304** method, the PWV can be quantified in various ways, including measurement of (i) a vessel's top-wall-displacement speed, (ii) a vessel's bottom-wall-displacement speed, or (iii) a rate of change of the vessel diameter (e.g., distance between the top wall and the bottom wall of the vessel).

[0036] As described herein, the top wall and bottom wall are defined relative to the ultrasound scanner **402**. For example, the “top” wall (e.g., top wall **412**) is the vessel wall nearest to the scanner **402** such that the top wall is the first surface of the vessel encountered by the ultrasound signals transmitted by the scanner **402**. In contrast, the “bottom” wall (e.g., bottom wall **414**) is the

vessel's opposing wall from the top wall and is farther from the scanner **402** than the top wall.
[0037] In a first example, only the top-wall-displacement speed is measured along the vessel. The ultrasound scanner **402** applies the M-line-1 **404** on the vessel **408** at an upstream location and the M-line-2 **406** at a downstream location relative to the direction of blood flow in the vessel **408**. The M-line-1 **404** passes through the vessel **408** at a first location **416** on the top wall **412** and the M-line-2 **406** passes through the vessel **408** at a second location **418** on the top wall **412**.
[0038] Plot **420** represents the displacement of the top wall **412** of the vessel **408** over time, where the displacement is caused by pulse waves pushing blood through the vessel **408**. The solid curve (e.g., curve **422**) represents the displacement over time of the top wall **412** of the vessel **408** at the first location **416**, as detected via the M-line-1 **404**. The dashed curve (curve **424**) represents the displacement over time of the top wall **412** of the vessel **408** at the second location **418**, as detected via the M-line-2 **406**. As the blood flows from left to right in the vessel **408** of the setup **400**, the vessel wall from M-line-1 **404** changes first due to the pulse wave, and the vessel wall from M-line-2 **406** changes at a later time. The time difference between the changes at the two locations (e.g., the first location **416** from M-line-1 **404** and the second location **418** from M-line-2 **406**) is a pulse propagation time (PPT), which can be calculated from a time shift between the two curves in the plot **420**. The time shift between the two curves can be determined using any suitable technique, including a cross-correlation method, a phase-shift calculation (e.g., time difference divided by wave period), transformation formula, etc. The cross-correlation method can determine the integral of the product of the two curves at multiple positions along the x-axis to find a maximum (or minimum) value of the product of the two curves. The determined maximum (or minimum) value indicates the two curves match. The time delay between the two curves can therefore be determined by the argument of the maximum (or minimum) of the cross-correlation. The PPT can be expressed as:
[00001] $PPT = \quad t + t_P \quad \mathrm{Eq.}(1)$
where Δt is the time shift from the two curves in the plot **420**, and t.sub.p is the time difference between the M-line-1 **404** and the M-line-2 **406**. Here, it is assumed that lines are fired alternately at positions of the M-line-1 **404** and the M-line-2 **406**. In an example, the M-line-1 **404** is fired first, then the M-line-2 **406** is subsequently fired, followed by another firing of M-line-1 **404**, and then again M-line-2 **406**. Such alternating firing of the M-lines can avoid mixing signals. Therefore, the PWV can be calculated as:
[00002] $PWV = \frac{d}{PPT \times \sin} \quad \mathrm{Eq.}(2)$
where θ is an angle between the blood vessel and the M-line and d is the distance (e.g., distance d.sub.1 **410**) between the M-lines. Because the distance d is predefined, its value is known for the calculations. If the vessel is parallel to the scanner, PWV can be reduced to:
[00003] $PWV = \frac{d}{PPT} \quad \mathrm{Eq.}(3)$
[0039] For a typical linear transducer, the transducer size is around 40 millimeters (mm). To measure the PWV of, for example, 10 m/s, the PPT is around 4 milliseconds (ms). To satisfy the Nyquist sampling criteria, the frame rate of each M-line needs to be about 1000 frames per second (fps). In aspects, the M-line-1 **404** and M-line-2 **406** can be acquired alternately to avoid mixing signals.
[0040] To increase the frame rate, an encoding and decoding mechanism can be used to fire both M-lines and acquire the data within a time interval (including a zero time interval, e.g., simultaneously). One example method is to use a bipolar Hadamard encoding and decoding processes for M-line-1 and M-line-2. For M-line-1, the signals generated with a Tx.sub.1 waveform can be described as S(M-line-1)=Rx.sub.1. For M-line-2, the signals generated with a Tx.sub.2 waveform can be described as S(M-line-2)=Rx.sub.2. If Tx.sub.1 an Tx.sub.2 are mixed together with both having positive polarity, the generated signals with this encoded waveform can be expressed as S(M-line-1+M−line-2), for example:

[00004] $S(M\text{-line-}1 + M\text{-line-}2) = Rx_1 + Rx_2 \quad \text{Eq.}(4)$

[0041] For simplicity, the first mixed transmit and receive event is referred to as Ping1. The received radio-frequency (RF) data of Ping1 RF.sub.Ping1 can be expressed as:

[00005] $RF_{\text{Ping1}} = RF_1 + RF_2 \quad \text{Eq.}(5)$

where RF.sub.1 is the RF data of M-line-1 from transmit Tx.sub.1, and RF.sub.2 is the RF data of M-line-2 from transmit Tx.sub.2. For a second mixed transmit and receive event, Ping2, Tx.sub.1 still has positive polarity while Tx.sub.2 has negative polarity. Combining these two, the transmit of Ping2 is Tx.sub.1+(−Tx.sub.2), and Tx.sub.1−Tx.sub.2 is used instead to represent the Ping2 transmit. Accordingly, the signals with this encoded waveform can be expressed as S(M-line-1-M−line-2):

[00006] $S(M\text{-line-}1 - M\text{-line-}2) = Rx_1 - Rx_2 \quad \text{Eq.}(6)$

[0042] The received RF data of Ping2 RF.sub.Ping2 can be expressed as:

[00007] $RF_{\text{Ping2}} = RF_1 - RF_2 \quad \text{Eq.}(7)$

[0043] Based on Eqs. (5) and (7), the RF data of M-line-1, which is RF.sub.1, and the RF data of M-line-2, which is RF.sub.2, can be recovered by using the following expressions:

[00008] $RF_1 = (RF_{\text{Ping1}} + RF_{\text{Ping2}})/2 \quad \text{Eq.}(8) \quad RF_2 = (RF_{\text{Ping1}} - RF_{\text{Ping2}})/2 \quad \text{Eq.}(9)$

where the decode Hadamard matrix has the same form of the encode matrix, which is:

[00009] $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

[0044] In this case, after using the encoding and decoding process, the M-line-1 and M-line-2 can be fired and acquired at the same time, which increases the frame rate by a factor of 2 and increases the range and accuracy of the PWV measurement. Note that other encoding methods can also be used, including unipolar Hadamard, Fourier, Wavelet, etc. The Hadamard matrix described above is exemplary and not meant to be limiting.

[0045] Although the above description uses the displacement of the top wall of the vessel from two M-lines to calculate PWV, the PWV can also be calculated using the displacement of the bottom wall of the vessel, as shown in plot **426**. For example, the calculation process is the same as using the top wall displacement of the vessel except the M-lines measure displacement at a third location **428** and a fourth location **430**, respectively, on the bottom wall **414** of the vessel **408**. Then, curves **432** and **434**, shown in the plot **426**, represent displacement over time of the bottom wall **414** at the third and fourth locations **428** and **430**, respectively.

[0046] In addition, using the same process, a change of distance between the top and bottom walls (e.g., vessel diameter **436**) can be measured and used to calculate the PWV. For example, M-line-1 **404** can be used to detect the change to the diameter **436** between the first location **416** on the top wall **412** and the third location **428** on the bottom wall **414**. The M-line-2 **406** can be used to detect the change to the diameter **436** between the second location **418** on the top wall **412** and the fourth location **430** on the bottom wall **414**. Then, the time difference (e.g., pulse propagation time) between when the changes occur along the M-line-1 **404** and when corresponding changes occur along the M-line-2 **406** is used to calculate the PWV, using Equations (1)-(9).

Dual PW (With or Without Encoding)

[0047] FIG. **5** illustrates an example implementation of measuring PWV using dual-PW Doppler (e.g., dual PW **306** in FIG. **3**). As mentioned, dual PW is another method usable to measure the rate of change of blood flow (e.g., blood-flow-change speed or blood-flow-velocity change). As shown in FIG. **5**, the principle and calculation processes are similar to the dual M-mode method described above, but the dual-PW method is different in that the blood flow is measured at upstream and downstream positions of a blood vessel. Then, by quantifying the blood-flow-change delays in these two positions, PWV can be calculated.

[0048] For example, setup **500** includes the ultrasound scanner(s) **402** applying two PW lines (e.g., PW-1 **502** and PW-2 **504**) with gates (e.g., sample volumes) positioned inside the same vessel (e.g., vessel **408**) at locations **506** and **508** that are separated by a distance d.sub.2 **510**. The two PW lines can be applied by the same scanner or two separate scanners. The distance d.sub.2 **510** between the PW lines is known. When applied by a single scanner, the distance d.sub.2 **510** is small (e.g., 5 mm, 3 mm, 1 mm, between 1 and 0.001 mm), such that the two PW lines are applied without physically moving the scanner **402**. The distance d.sub.2 **510** can be greater when applied by separate scanners; however, a greater distance d.sub.2 **510** can be less accurately defined and can therefore reduce the accuracy of the PWV calculations. The PW-1 **502** line corresponds to a first location **506** in the vessel **408** and the PW2 **504** line corresponds to a second location **508** in the vessel **408**.

[0049] Plot **512** represents blood-flow velocity measured over time. The solid curve (e.g., curve **514**) is blood-flow velocity measured from the PW-1 line (e.g., PW-1 **502**). The dashed curve (e.g., curve **516**) is blood-flow velocity measured from the PW-2 line (e.g., PW-2 **504**). As blood flows from left to right in the vessel **408** of the setup **500**, the blood-flow velocity at PW-1 **502** changes first due to the pulse wave, and the blood-flow velocity at PW-2 **504** changes later. The time difference between the blood-flow-velocity change at the two locations (e.g., the first location **506** and the second location **508**) represents the pulse propagation time (PPT), which can be calculated from the time shift between the two curves in plot **512** (e.g., curve **514** and curve **516**). The time shift can be calculated based on a cross-correlation method applied to the two curves in plot **512**. Such calculations are based on equations similar to those described above for the dual M-mode example, including Eqs. (1)-(9), where the distance d.sub.2 **510** is used as the distance d in the calculations. In addition, a similar encoding and decoding method can also be applied.

[0050] The vessel-wall movement (described with respect to FIG. **4**) and the blood-flow-velocity change (described with respect to FIG. **5**) are the two results caused by a pulse wave. Therefore, if both the vessel-wall movement and the blood-flow-velocity change are measured, a more accurate measurement of the PWV can be achieved. One way to measure both the vessel-wall movement and the blood-flow-velocity change is to use dual PW only, as described herein with respect to FIG. **6**. Another method is to combine the dual M-mode **304** and dual PW **306** methods, which is described with respect to FIG. **7** in more detail.

[0051] FIG. **6** illustrates an example implementation of using a dual-PW method (e.g., dual PW **306** in FIG. **3**) to measure both blood-flow-velocity changes and vessel-wall movement. Although the PW waveform usually is long in order to have a narrow bandwidth and better flow measurement sensitivity, the PW waveform is still a pulse. Accordingly, prior to performing the PW-blood-flow process, a typical beamforming method can be implemented to form a line image (similar to M-mode) and thus calculate the vessel-wall displacement and eventually vessel diameter. The spatial resolution may be low in the dual PW method, but such spatial resolution is sufficient to measure the vessel diameter and thus measure the PWV.

[0052] As shown in FIG. **6**, setup **600** is the same as the setup **500** in FIG. **5**. However, in addition to measuring the blood-flow-velocity changes over time (as shown in plot **512**) using the dual PW method described above, displacement of the blood vessel's top wall **412** and bottom wall **414** can also be extracted (similar to plots **420** and **426** in FIG. **4**). Then, based on locations associated with the top and bottom walls (e.g., the first location **416**, the second location **418**, the third location **428**, and the fourth location **430**), the vessel diameter **436** can be calculated, as shown in plot **602**. The solid line (e.g., curve **604**) represents the vessel diameter **436** as calculated using the measurements extracted from the first location **416** and the third location **428** using PW-1 **502**. The dashed line (e.g., curve **606**) represents the vessel diameter **436** as calculated using the measurements extracted from the second location **418** and the fourth location **430** using PW-2 **504**.

[0053] Note that although the illustrated example in FIG. **6** shows vessel-diameter measurements in plot **602**, it is enough to measure PWV just based on the vessel's top-wall displacement or bottom-wall displacement, as shown and described with respect to FIG. **4** (e.g., plots **420** and **426**).

Combination Mode

[0054] FIG. **7** illustrates an example implementation of using a combination mode to measure PWV. In particular, the illustrated example uses a combination of the dual M-mode **304** and the dual PW **306** to measure PWV. A setup **700** for measurement includes two M-lines (e.g., M-line-1 **404**, M-line-2 **406**) used to measure the same vessel and two PW lines (e.g., PW-1 **502**, PW-2 **504**) with respective gates inside the vessel and used to measure the same vessel. An M-line and a PW line can be aligned (same vertical line) or have different positions. For simplicity, the example described in FIG. **7** shows M-lines placed at the same positions as the PW lines. However, the techniques described herein can also be implemented with the M-lines being offset (e.g., unaligned) from the PW lines. In the example illustration, the distance d.sub.1 **410** between the M-mode lines is the same as the distance d.sub.2 **510** between the PW lines. In practice, the distances d.sub.1 and d.sub.2 are not required to be the same, as long as the distances d.sub.1 and d.sub.2 are accurately determined when used to calculate the PWV, as indicated in Equation (3).

[0055] Using dual M-mode together with dual PW can enhance the spatial resolution. In this case, if dual M-mode lines and dual-PW lines are fired alternately (continually following and succeeded by each other, one after the other, e.g., interleaved), then the frame rate becomes half with respect to the dual M-mode or the dual PW, which can limit the maximum measurable PWV to be half. However, because M-mode has better spatial resolution than PW, the measurement accuracy from wall displacement and diameter of the blood vessel is expected to be better than the result from dual PW.

[0056] The change in blood-flow velocity over time is illustrated in plot **512** and is measured using the techniques described herein with respect to FIG. **5**. For example, the solid curve (e.g., curve **514**) is blood-flow velocity measured from PW-1 **502** at the first location **506** and the dashed curve (e.g., curve **516**) is blood-flow velocity measured from PW-2 **504** at the second location **508**. As disclosed above, the time difference between the blood-flow-velocity changes at these two locations represents the pulse propagation time (PPT), which can be calculated from the time shift between the two curves in plot **512** (e.g., curve **514** and curve **516**). The time shift can be calculated based on a cross-correlation method applied to the two curves in plot **512**.

[0057] The change in vessel diameter over time is illustrated in plot **702**. In plot **702**, the solid line (e.g., curve **704**) represents changes to the vessel diameter **436** over time as determined using the M-line-1 **404** to measure vessel-wall displacement of the top and bottom walls of the vessel **408**, as measured at the first location **416** on the top wall **412** and the third location **428** on the bottom wall **414**. The dashed line (e.g., curve **706**) represents the changes to the vessel diameter **436** over time as determined using the M-line-2 **406** to measure vessel-wall displacement of the vessel **408**, as measured at the second location **418** on the top wall **412** and the fourth location **430** on the bottom wall **414**. In some implementations, however, it is sufficient to measure PWV just based on the displacement of the top wall **412** of the vessel **408** or the displacement of the bottom wall **414** of the vessel **408**, as described with respect to FIG. **4**.

[0058] To increase the frame rate, similar encoding and decoding methods can be used on dual M-mode and dual PW. Instead of encoding two modes, four modes are encoded (two M-mode lines and two PW lines). Therefore, a Hadamard bipolar method of order four can be used. In this case, the frame rate can be increased by a factor of four. Other similar encoding methods can also be used, including unipolar Hadamard, Fourier, Wavelet, etc.

[0059] To calculate PWV, the ultrasound machine **102** measures vessel-wall movement and blood-flow change. Therefore, any ultrasound modes that can be used to measure vessel-wall movement and blood-flow change are applicable. For example, the ultrasound machine **102** can use (i) a high frame rate B-mode image, or anatomic M-mode, to measure the vessel-wall movement and (ii) high frame rate color flow imaging to measure the change in blood-flow velocity. Such a procedure is similar to the dual M-mode **304** or the dual PW **306**, which are used to measure one property (vessel-wall movement or blood flow) at an upstream location and the same property (vessel-wall movement or blood flow) at a downstream location of the same vessel. Then based on a correlation method, the PPT is calculated and used to calculate PWV with the known distance between the two locations.

Blood Pressure Measurement

[0060] After successfully obtaining a PWV measurement, the blood pressure (BP) can be estimated based on the Moens-Koreweg (MK) and Hughes equations:

[00010] $PWV = \sqrt{\frac{Eh_0}{2\ R_0}}$ Eq. (10) $E = E_0 \exp(\ BP)$ Eq. (11)

where E is the elastic modulus at blood pressure BP, h.sub.0 is the thickness of the vessel (e.g., artery), and R.sub.0 is the radius of the vessel. In addition, ρ is the blood density, E.sub.0 is the elastic modulus at zero blood pressure, and τ is a material coefficient of the vessel. As shown in Equations (10) and (11), as the blood pressure BP increases, the PWV increases as well. The relationship between BP and PWV can be expressed as follows:

[00011] $BP = \frac{1}{\ }\ln\left(\frac{4(\ R_0 PWV)^2}{h_0 E_0}\right)$ Eq. (12)

[0061] Using Equation (12), the blood pressure BP can be determined based on the PWV. One technique to measure BP based on PWV is to use an empirical method, which is first to measure a series of patients with different PWVs and BPs, and then based on Eq. (12), perform a regression to determine the coefficients between PWV and BP. In that case, a universal empirical formula is established and can be used to predict BP based on PWV in a future patient. Another technique to measure BP based on PWV is to establish a formula based on one patient's data and use that formula for future measurements of other patients. In some implementations, machine learning (ML) can be used to determine BP based on PWV. An ML model can ignore the above-described equations. For example, an ML model is established based on a collection of data, such as measurements from a series of patients with different PWVs and BPs, and then the ML model is used to predict the BP in a new patent based on a measured PWV of that new patient.

[0062] By using the techniques disclosed herein (dual M-mode, dual PW, dual M-mode+dual PW, etc.), other important physiological properties can also be measured with increased accuracy. For example, M-mode and PW can be used at the same blood-vessel position to measure blood volume in real time with high accuracy. Blood volume per second (BVS) can be calculated as:

[00012] $BVS = \frac{\ d^2 V}{4}$ Eq. (13)

where d is the diameter of the vessel and V is the blood-flow velocity. Compared to conventional methods for measuring BVS, which are generally based on a B-mode image to measure vessel diameter, the disclosed techniques can provide significantly higher temporal resolution (e.g., frame rate) for BVS measurement. A fast frame rate is critical for various applications, including cardiac function monitoring. For example, the M-line and PW gate can both be placed on a valve, such as a mitral valve, a tricuspid valve, a pulmonary valve, or an aortic valve. Then, a corresponding ejection fraction can be measured with a high frame rate and high accuracy.

[0063] Another application of using M-mode and PW to monitor the same location on a vessel is to quantify the phase delay of a pulse wave along the vessel wall and the blood. Although the pulse-wave source is the same (e.g., from the heart), the propagation along the vessel wall and the blood stream depend on other physiological properties, such as vessel-wall stiffness and blood viscosity. Therefore, by monitoring the delays of the pulse wave along the vessel wall and inside the blood stream, the vessel-wall stiffness and the blood viscosity can each be calculated.

[0064] Dual M-mode and dual PW can also be used to measure the blood volume at upstream and downstream locations in the same vessel at the same time. Not only can PWV be measured, but these techniques can also be used to calculate conservation of the blood volume by comparing the blood-volume-measurement results from the two locations. Then, based on the comparison of the blood-volume-measurement results, the ultrasound system can determine potential leakage or blockage of the blood vessel.

[0065] Other implementations can include using M-mode and PW on the same position. For example, the relative phase shift between vessel-wall-displacement speed and blood-velocity changes can indicate one or more physiological properties of the heart and of the vessel between the measurement point and the heart. M-mode and PW can be used on the same position (valve, artery, etc.) to measure the vessel-diameter-change speed and the blood-flow velocity at the same time, which can enable blood flux to be quantitatively monitored in real time.
[0066] By using M-mode and PW on two positions, such as the example described with respect to FIG. **7**, a difference in blood flux at the two positions can be measured. The difference in blood flux at the two positions can be used to determine blood perfusion and blood-perfusion indexes.
[0067] Accordingly, different ultrasound-imaging modes can be combined to maximize the capability of the modes used. For example, for PWV measurement, a higher pulse-repetition frequency (PRF) of M-mode and/or PW provides enhanced measurement results.
Example Methods
[0068] FIGS. **8** and **9** depict methods **800** and **900**, respectively, for using ultrasound to measure physiological properties. The methods **800** and **900** are shown as a set of blocks that specify operations performed but are not necessarily limited to the order or combinations shown for performing the operations by the respective blocks. Further, any of one or more of the operations can be repeated, combined, reorganized, or linked to provide a wide array of additional and/or alternate methods. In portions of the following discussion, reference can be made to the example system **100** of FIG. **1** or to entities or processes as detailed in FIGS. **2-7**, reference to which is made for example only. The techniques are not limited to performance by one entity or multiple entities operating on one device.
[0069] FIG. **8** depicts a method **800** for using ultrasound to measure physiological properties. The method **800** can be performed by the ultrasound system **100**. At **802**, a multi-mode ultrasound is applied to acquire ultrasound data at two locations of a same vessel that are separated by a known distance. For example, multiple ultrasound modes (e.g., dual M-mode **304**, dual PW **306**, combination mode **308**) can be used by the ultrasound control subsystem **218** to process the raw ultrasound data associated with received signals from the scanner **402**, where the received signals are received from two locations separated by a known distance.
[0070] At **804**, one or more characteristics of the vessel is measured at each of the two locations. For example, the ultrasound machine **102** can measure one or more characteristics of the vessel **408** at each of the two locations, including vessel-wall-displacement speed, blood-flow-change speed, etc.
[0071] At **806**, a phase delay of a pulse wave along the vessel between the two locations is quantified based on a correlation between the one or more characteristics of the vessel at the two locations. For example, the one or more characteristics (vessel-wall-displacement speed, blood-flow-change speed, etc.) are used to determine the pulse propagation time (PPT) of the pulse wave propagating through the vessel **408**.
[0072] At **808**, a physiological property associated with the vessel is calculated based on the known distance and the phase delay. For example, the distance d and the PPT can be used to calculate the PWV. In aspects, the PWV can be calculated by the ultrasound system **100** using at least some of the Eqs. (2)-(9) disclosed above.
[0073] In some aspects, at **810**, one or more additional physiological properties associated with the vessel are determined based on the calculated physiological property. For example, the calculated physiological property (e.g., PWV) can be used to calculate an additional physiological property, such as BP, associated with the vessel **408**.
[0074] FIG. **9** depicts a method **900** for measuring physiological properties using ultrasound. The method **900** can be implemented by the ultrasound system **100**. At **902**, first ultrasound data corresponding to a first location on a vessel is acquired over a duration of time. In one example, the M-line-1 **404** is applied to an upstream location (e.g., the first location **416**, the third location **428**).

In another example, the PW-1 **502** line is applied to the first location **506**.

[0075] At **904**, second ultrasound data corresponding to a second location on the vessel is acquired over the duration of time, where the second location is separated from the first location by a known distance. In one example, the M-line-2 **406** is applied to a downstream location (e.g., the second location **418**, the fourth location **430**), where the downstream location is separated from the upstream location by the distance d.sub.1 **410**. In another example, the PW-2 **504** gate is applied to the second location **508**, where the second location is separated from the first location **506** by the distance d.sub.2 **510**.

[0076] At **906**, at least one of a rate of vessel-wall displacement of the vessel or blood-flow-velocity changes in the vessel is measured over the duration of time based on the first ultrasound data and the second ultrasound data. For example, vessel-wall-displacement speed can be measured using the M-mode lines. Alternatively, the blood-flow-velocity changes can be measured using the PW lines.

[0077] At **908**, a pulse-propagation time of a pulse wave along the vessel is calculated based on a correlation of the at least one of the vessel-wall displacement or the blood-flow-velocity changes between the first and second locations. For example, the pulse-propagation time of a pulse wave propagating through the vessel **408** can be calculated based on a phase delay of the vessel-wall-displacement speed between the first location **416** and the second location **418**, or between the third location **428** and the fourth location **430**. In another example, the pulse-propagation time of a pulse wave propagating through the vessel **408** can be calculated based on a phase delay of the blood-flow-velocity changes between the first location **506** and the second location **508**.

[0078] At **910**, a pulse-wave velocity of blood in the vessel is determined based on the pulse-propagation time and the known distance. For example, the ultrasound system **100** can use Equations (1)-(9) to determine the PWV.

[0079] In some aspects, at **912**, one or more physiological properties associated with the vessel is determined based on the pulse-wave velocity. For example, blood pressure can be determined based on the pulse-wave velocity. Other physiological properties can be determined, including vessel-wall stiffness, blood viscosity, blood volume, blood perfusion, blood-perfusion indexes, and so on.

Example Machine-Learned Models

[0080] FIG. **10** represents an example ML model for processing an input from an ultrasound device (e.g., the ultrasound machine **102**). In aspects, an ML model **1000** can include inputs **1002**, a neural network **1004**, and a generation module **1006**. The inputs **1002** can, by way of example, include ultrasound data, which can be used to generate ultrasound images **1008** (e.g., first M-mode image **1008-1**, second M-mode image **1008-2**) and a PWV **1010** corresponding to the ultrasound data and/or the ultrasound images **1008**. Additionally or alternately, the ultrasound images **1008** and the PWV **1010** can be used as the inputs **1002** for the ML model **1000** to provide an output **1012**.

[0081] According to some implementations, the generation module **1006** can comprise a segmentation head **1014**, an object detection head **1016**, a classification head **1018**, or more or fewer components. The segmentation head **1014** can, in some examples, highlight structures of interest in the inputs **1002**, such as by generating segmentations of the structures and/or segmentation images **1020**. According to some embodiments, the object detection head **1016** can generate bounding boxes **1022** of structures of interest in the inputs **1002**. The classification head **1018** can, for example, determine physiological properties **1024** from the inputs **1002**. The physiological properties are properties associated with the vessel, and can include blood pressure **1026**, arterial stiffness **1028**, blood volume **1030**, blockage/leakage **1032**, etc. Using such an ML model can increase the accuracy of the determined physiological properties in comparison to the approximations calculated using the deterministic approach (e.g., Equations (10)-(13)).

[0082] The inputs **1002** can, in some implementations, be input as data, such as a matrix or other, multi-dimensional mathematical object, the ultrasound data generated by the ultrasound scanner

104, and so on. In aspects, the neural network **1004** can include a feature-extraction component, such as a convolutional neural network (CNN). The neural network **1004** can, in some examples, comprise several different neural network architectures known to a person of ordinary skill in the art and can comprise any combination of like or different architectures. According to some embodiments, the neural network **1004** can comprise a single architecture type. These example neural network architectures and combinations are listed as examples only and are not meant to limit the scope of the neural network **1004**. It should be noted that the neural network **1004**, for example, can comprise a network other than a learning network, as can be construed by the term "neural network." Rather, in aspects the neural network **1004** can comprise an algorithm derived from a machine-learning training, as is explained below.

[0083] Many of the aspects described herein can be implemented using an ML model. For the purposes of this disclosure, an ML model is any model that accepts an input, analyzes and/or processes the input based on an algorithm derived via machine-learning training, and provides an output. An ML model can be conceptualized as a mathematical function of the following form:

[00013] $f(\hat{s},\quad) = \hat{y}$ Equation(14)

[0084] In Equation (14), the operator f can represent the processing of the ML model based on an input and providing an output. The term ŝ can represent a model input, such as ultrasound data, optical data, or both or other data. The ML model can analyze/process the input ŝ using parameters θ to generate an output ŷ (e.g., object identification, object segmentation, object classification). Both the input ŝ and the output ŷ can be scalar values, matrices, vectors, or mathematical representations of phenomena such as categories, classifications, image characteristics, the images themselves (e.g., the ultrasound image **116**), text, labels, or the like. The parameters θ can be any suitable mathematical operations, including but not limited to applications of weights and biases, filter coefficients, summations or other aggregations of data inputs, distribution parameters such as mean and variance in a Gaussian distribution, linear algebra-based operators, or other parameters, including combinations of different parameters, suitable to map data to a desired output.

[0085] FIG. **11** represents an example machine-learning architecture **1100** used to train an ML model **1102** (e.g., ML model **1000**). An input module **1104** can accept an input ŝ **1106**, which can be an array with members ŝ.sub.1 through ŝ.sub.n. The members of the array can be multidimensional, or the array itself can instead be a matrix or other mathematical object holding multiple points or vectors of data values. The input ŝ **1106** can be fed into a training module **1108**, which can process the input ŝ **1106** based on the machine-learning architecture **1100**. For example, if the machine-learning architecture **1100** uses a multilayer perceptron (MLP) model **1110**, the training module **1108** applies weights and biases to the input ŝ **1106** through one or more layers of perceptrons, each perceptron performing a fit using its own weights and biases according to its given functional form. The MLP weights and biases can be adjusted such that they are optimized against a least mean square, logcosh, or other optimization function (e.g., loss function) known in the art. Although the MLP model **1110** is described here as an example, any suitable machine-learning technique can be employed, some examples of which include but are not limited to k-means clustering **1112**, convolutional neural networks (CNN) **1114**, a Boltzmann machine **1116**, a Gaussian mixture model (GMM), and a long short-term memory (LSTM). The training module **1108** can provide an input to an output module **1118**. The output module **1118** can analyze the input from the training module **1108** and provide an output in the form of ŷ **1120**, which can be an array with members ŷ.sub.1 through ŷ.sub.m, or another single-or multiple-dimensional object. The output ŷ **1120** can represent a known correlation with the input ŝ **1106**, such as, for example, object identification, segmentation, and/or classification.

[0086] In some examples, the input {circumflex over (**3**)} **1106** can be a training input labeled with known output correlation values, and these known values can be used to optimize the output ŷ **1120** in training against the optimization/loss function. In other examples, the machine-learning architecture **1100** can categorize the output ŷ **1120** values without being given known correlation values to the inputs ŝ **1106**. In some examples, the machine-learning architecture **1100** can be a combination of machine-learning architectures. By way of example, a first network can use the input ŝ **1106** and provide the output ŷ **1120** as an input ŝ.sub.ML to a second machine-learned architecture, with the second machine-learned architecture providing a final output ŷ.sub.f. In another example, one or more machine-learning architectures can be implemented at various points throughout the training module **1108**.

[0087] In some machine-learned models, all layers of the model can be fully connected. For example, all perceptrons in the MLP model **1110** act on every member of ŝ. For the MLP model **1110** with a 100×100 pixel image as an input, each perceptron provides weights/biases for 10,000 inputs. With a large, densely layered model, this can result in slower processing and/or issues with vanishing and/or exploding gradients. The CNN **1114**, which can, in some constructions, not be a fully connected model, can process the same image using 5×5 tiled regions, requiring only 25 perceptrons with shared weights, giving much greater efficiency than the fully connected MLP model **1110**. Additionally or alternately, the training module **1108** can employ sections of architecture that are fully connected and sections that are not. By way of example, the input ŝ **1106** can be a matrix representing data from both an ultrasound image and an optical image and the training module **1108** can use the CNN **1114** to identify features from the input ŝ **1106**. The features can then be used as auxiliary inputs for the MLP model **1110**, which can subsequently give an output to the output module **1118**. The CNN **1114** portion of this example is not fully connected, but the MLP **1110** portion is fully connected, where every perceptron in a first layer takes every input from the CNN **1114** and all perceptrons are causally connected to the eventual input for the output module **1118**. Other architecture types and combinations can be employed by a person of ordinary skill in the art, and the foregoing example is not meant to be limiting, but illustrative.

[0088] FIG. **12** represents an example model **1200** using a CNN to process an input image **1202**, which can include representations of objects that can be identified via object recognition, such as people or cars (or an anatomy, such as the vessel **408**). Convolution A **1204** can be performed, for example, to create a first set of feature maps (e.g., feature maps A **1206**). A feature map can be a mapping of aspects of the input image **1202** given by a filter element of the CNN. This process can be repeated using, by way of example, feature maps A **1206** to generate further feature maps B **1208**, feature maps C **1210**, and feature maps D **1212** using convolution B **1214**, convolution C **1216**, and convolution D **1218**, respectively. In this example, feature maps D **1212** can become an input for fully connected network layers **1220**. In this way, the example model **1200** can be trained to recognize certain elements of the image, such as people, cars, or a particular patient anatomy, and provide an output **1222** (a prediction, an inference, etc.) that, for example, can identify the recognized elements. Additionally or alternately, each feature map, such as feature maps C **1210**, can contain multiple feature maps. It is possible for some feature maps to be a set of multiple feature maps and others to be a single feature map. By way of example, feature maps A **1206** can be multiple feature maps, each using a variation of the CNN architecture of the example model **1200**, and feature maps B **1208** can be a single feature map.

[0089] Although the example of FIG. **12** shows a CNN as a part of a fully connected network, other architectures are possible, and this example should not be seen as limiting. There can be more or fewer layers in the CNN. The CNN component for the model can be placed in a different order, or the model can contain additional components or models. In some examples, there are no fully connected components, such as in a fully convolutional network. Additional aspects of the CNN, such as pooling, down-sampling, up-sampling, or other aspects known to a person of ordinary skill in the art, can also be employed.

[0090] FIG. **13** illustrates an example user interface **1300** displayed via a display device (e.g., the display device **108**) coupled to the ultrasound system **100** from FIG. **1**, in accordance with one or more implementations. In aspects, the user interface **1300** can be used to provide results of multiple different ultrasound imaging modes, including those described with respect to FIGS. **3-12**. In the illustrated example, the user interface **1300** includes a first portion **1302**, a second portion **1304**, and a third portion **1306**. The user interface **1300** can include additional (or fewer) portions for presenting ultrasound information.

[0091] In an example, the first portion **1302** of the user interface **1300** can be used to present an ultrasound image **1308** generated by the ultrasound system **100** using B-mode plus color Doppler. The color Doppler can be superimposed or overlaid over the B-mode image. For example, a section **1310** of the B-mode image (e.g., the ultrasound image **1308**) can be selected for color Doppler analysis. In FIG. **13**, dotted areas **1312** represent colored areas displayed over the B-mode image. The dotted areas **1312** are colored with a color that represents fluid flow in a direction. A different color can be used to represent fluid flow in an opposing direction (e.g., red for a first flow direction and blue for a second, opposite flow direction).

[0092] The second portion **1304** of the user interface **1300** can be used to present results associated with PW Doppler measurements corresponding to a PW line **1314** having a gate **1316** selected or placed in the ultrasound image **1308** in the first portion **1302** of the user interface **1300**. The third portion **1306** of the user interface **1300** can be used to present results associated with M-mode ultrasound corresponding to one or more M-lines selected or placed in the ultrasound image **1308** in the first portion **1302** of the user interface **1300**. In the illustrated example, the M-line is collocated with the PW line **1314**. However, the M-line can be placed at a different location than the PW line **1314**. Although only one PW line **1314** is shown, multiple PW lines **1314** and/or M-lines can be used, as described with respect to FIGS. **5-7**, to determine a physiological property such as PWV.

[0093] Accordingly, the user interface **1300** can be used to provide information corresponding to a plurality of ultrasound modes simultaneously. In addition to providing the ultrasound mode information (e.g., the ultrasound image **1308**, the PW results, the M-mode results, etc.), the user interface **1300** can also provide an indication of one or more physiological properties **1318**, as disclosed above. The physiological properties, as determined from the ultrasound data associated with the ultrasound image **1308**, can include PWV, blood pressure, arterial stiffness, blood volume, blockage/leakage percentage, etc.

CONCLUSION

[0094] Ultrasound methods and systems for measuring physiological properties are disclosed. These ultrasound techniques provide highly accurate measurements of physiological properties of a subject by reducing errors associated with distance estimations between the measurement location and the heart. Additionally, these ultrasound techniques provide a way to measure physiological properties, such as pulse-wave velocity and blood pressure, using only ultrasound and without relying on the assistance of ECG data.

Claims

1. An ultrasound system comprising: an ultrasound scanner configured to generate ultrasound data based on reflections of ultrasound signals transmitted by the ultrasound scanner at an anatomy, the ultrasound scanner configured to acquire the ultrasound data at two locations of a same vessel that are separated by a distance, the ultrasound data acquired at the two locations within a time interval; one or more processors; and one or more computer-readable storage media having instructions stored thereon that, responsive to execution by the one or more processors, cause the one or more processors to: determine one or more characteristics of the vessel at each of the two locations using the ultrasound data; determine, based on a correlation between the one or more characteristics of the vessel at the two locations, a phase delay of a pulse wave propagating through the vessel between the two locations; and determine a physiological property associated with the vessel based on the one or more characteristics and the distance.

2. The ultrasound system of claim 1, wherein the physiological property is a pulse-wave velocity of the pulse wave.
3. The ultrasound system of claim 1, wherein the one or more characteristics include at least one of a vessel-wall displacement over time, blood-flow-velocity changes over time, or vessel-diameter-change speed over time.
4. The ultrasound system of claim 1, wherein the instructions further cause the one or more processors to determine, based on the physiological property, one or more additional physiological properties associated with the vessel.
5. The ultrasound system of claim 4, wherein the one or more additional physiological properties include at least one of blood pressure, blood volume per second, vessel-wall stiffness, or blood viscosity.
6. The ultrasound system of claim 1, wherein the one or more characteristics of the vessel are measured without using electrocardiogram (ECG) data.
7. The ultrasound system of claim 1, wherein the ultrasound scanner is configured to acquire the ultrasound data at the two locations using dual M-mode or dual-PW Doppler.
8. The ultrasound system of claim 1, further comprising determining the distance that separates the two locations based on transducer elements used to generate the ultrasound signals at each of the two locations.
9. The ultrasound system of claim 1, wherein the ultrasound scanner is configured to acquire the ultrasound data at the two locations using two or more modes including two or more of an M-mode, a PW Doppler, a color Doppler, and a B-mode.
10. The ultrasound system of claim 1, wherein the ultrasound scanner is a single scanner configured to transmit the ultrasound signals at the two locations simultaneously.
11. The ultrasound system of claim 1, wherein the ultrasound scanner is a single scanner and transmission of the ultrasound signals is interleaved between the two locations.
12. A method comprising: acquiring ultrasound data at two locations of a same vessel that are separated by a distance, the ultrasound data acquired at the two locations within a time interval; determining one or more characteristics of the vessel at each of the two locations using the ultrasound data; determining, based on a correlation between the one or more characteristics of the vessel at the two locations, a phase delay of a pulse wave propagating through the vessel between the two locations; and determining a physiological property associated with the vessel based on the one or more characteristics and the distance.
13. The method of claim 12, wherein determining the physiological property includes calculating a pulse-wave velocity of the pulse wave propagating through the vessel.
14. The method of claim 12, wherein determining one or more characteristics of the vessel includes measuring at least one of a vessel-wall displacement over time, blood-flow-velocity changes over time, or vessel-diameter-change speed over time.
15. The method of claim 12, further comprising determining, based on the physiological property, one or more additional physiological properties associated with the vessel.
16. The method of claim 15, wherein determining the one or more additional physiological properties includes determining at least one of blood pressure, blood volume per second, vessel-wall stiffness, or blood viscosity.
17. The method of claim 12, wherein acquiring the ultrasound data at the two locations includes using dual M-mode ultrasound by applying a first M-line at a first location of the two locations and a second M-line at a second location of the two locations.
18. The method of claim 12, wherein acquiring the ultrasound data at the two locations includes using dual-PW Doppler by applying a first PW line at a first location of the two locations and a second PW line at a second location of the two locations.
19. The method of claim 12, wherein acquiring the ultrasound data at the two locations includes using two or more of an M-mode, a PW Doppler, a color Doppler, and a B-mode.

20. The method of claim 12, wherein acquiring the ultrasound data includes using a single scanner configured to transmit the ultrasound signals at the two locations simultaneously.